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Implementation of Sulphur Recovery and Carbon Capture Units to Simultaneously Unlock Production Potential as well as Reduce Overall Carbon Footprint in a Large Non- Associated Gas Condensate Field

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Abstract

A large non-associated gas condensate field consisting of two reservoirs is currently under recycle mode producing mostly condensate (C5+) and natural gas liquids (NGL). The produced gas from the field consists of approximately 3-4 % sour gas (H₂S) and approximately 4-5 % CO₂ which make conventional processing of the produced gas challenging. In this work, it is proposed to implement a Sulphur Recovery Unit (SRU) and a Carbon Capture Unit (CCU) unit at the processing plant and extract the sulphur from the H₂S gas and remove the CO₂ instead re-injecting the excess sour gas components back into the reservoir

In the initial step of the processing of the feed gas, the heavy components (C5+) are separated out from the produced gas in a Condensate Recovery Unit (CRU). The dry sour gas is then routed to an Acid Gas Removal Unit (AGRU) where the acid gas components (H₂S and CO₂) are completely removed from the feed stream. Finally, the intermediate components (C₂ to C₄) are extracted away from the sweet gas in the Natural Gas Liquids (NGL) unit. The lean gas (mostly C₁) from the effluent of the NGL unit is re-injected back into the reservoir. The acid gas components (H₂S and CO₂) stripped off at the processing plant are injected back in their entirety into the reservoir as there is no mechanism to safely dispose of the sour gas components. The total re-injected gas is maintained at a 75% recycle ratio to support the reservoir pressure. The re- injection of the sour components led to a gradual increase in the concentration of H₂S and CO₂ in the reservoirs. The increased acid gas concentrations were deemed detrimental to the facilities as the acid gas concentrations in the injection streams were forecasted to approach design limitations of the injection facilities in a few years based on current operations. The SRU unit consists of a catalytic unit with a capacity to remove up to 15 tons of elemental sulphur. The CCU unit consists of DMEA absorbent based technology.

The planned installation of the SRU and the CCU will gradually lower the concentrations of the acid gas components in the injection streams which will help in safe operation of the facilities. This implementation also helps in unlocking the production potential of the reservoir by enabling the ramp-up of the production from the field by approximately 10- 12 %. The elemental sulphur produced would generate an additional revenue stream. The captured CO₂ will be sent to a nearby oil field for enhanced oil recovery, adding to the existing revenue streams. In this paper we describe the process flow, design criteria and other aspects

such as operational challenges involved in the successful installation, implementation and commissioning of the SRU and CCU units.

Field Background

The subject field is located in the onshore area of Abu Dhabi about 110 km to the Southwest of Abu Dhabi Island. The field was discovered in 1965, with the first oil production starting in 1973. The field is an elongated, faulted, double plunging anticline with the longer axis trending roughly Northeast-Southwest. It is about 30 km long and 10 km wide.

General Geology

The field is an elongated faulted anticline with the longer axis trending roughly NE-SW. It is about 30 km long and 10 km wide. The field is one of a sequence of anticlines on the eastern margin of the Falaha Syncline. These were formed in the Late Cretaceous by the compressional effects of the collision of Oman with Arabia. Subsequent modification of the structure during the Tertiary has occurred due to burial compaction and also due to the Zagros orogeny which has regionally tilted the area down to the NNE. The symmetrical anticline is diagonally dissected by NW-SE trending possible wrench faults.

The field consists of two main gas bearing reservoirs belonging to the early Cretaceous Thamama group. Reservoirs R1 and R2 are part of the Lekhwair formation, a sequence of clean limestone reservoir horizons alternating with argillaceous limestone, the non-reservoir intervals are commonly referred as dense zones.

Five lithofacies have been recognized within the succession in R1 and R2 reservoirs, corresponding to open platform shoal grainstones (LF-4), inner ramp wackestones to rudstones (LF-5, LF-3, LF-2) and mid ramp mudstones to wackestones (LF- 1). Of these lithofacies, LF-3 and LF-5 dominate reservoir R1 and LF-2 and LF-3 dominate reservoir R2.

The R1 reservoir is thicker and of better reservoir quality than the R2 reservoir. The field contains some moderate throw faults, and a down dip gas-water contact at about 9235 ft TVDSS. Reservoir R1 thickness ranges from 53 ft down flank to 71 ft on the crest. Its lithology consists mainly of pelletal grainstone to packstone in the upper part and medium to fine grained packstone to wackestone in the lower part deposited as moderate to high energy open platform (shoal to middle ramp) processes. The average thickness of reservoir R1 is 60 ft.

The thickness of reservoir R2 ranges from 19 ft down flank to 30 ft on the crest. Its lithology consists mainly of packstone to pelletal grainstone with local wackestone in the upper part. The lower part is wackestone to packstone with local lime mudstone deposited by low to moderate energy subtidal platform. The average thickness of reservoir R2 is 25 ft.

The average reservoir porosity in R1 is 23% and the permeability is around 23 mD. Reservoir R2 has poor reservoir quality with average porosity of 13% and averaged permeability of 4 mD. A snapshot of the two reservoirs in terms of reservoir quality and thickness is shown in [Table 1](#).

Table 1—Lithology and Geological Properties of the two reservoirs

Reservoir	Unit	R1	R2
Reservoir Quality		Better	Poor
Average Reservoir Thickness	[ft]	60	25
Average Porosity	[%]	23	14
Average Permeability	[mD]	23	4
Depositional Environment		Moderate to high energy open platform (shoal to middle ramp)	Low to moderate energy subtidal platform

Reservoir	Unit	R1	R2
Lithology	Upper Part	Mainly pack to rudsntone and minor grainstone	Mainly packstone to pelletal grainstone with local wackestone
	Lower Part	Medium to ingrained packstone to wackestone	Wackestone to packstone with local musdstone
Reservoir Quality	Upper Part	Good	Poor
	Lower Part	Poor	Moderate

Fluid Properties

There are 4 initial samples available for reservoir R1 and two initial samples for reservoir R2. All the samples are recombined separator samples. [Figure 1](#) shows the top structure map, the location of all the producers and injectors and the locations of the sampled wells for reservoir R1 and [Figure 2](#) show the similar map for reservoir R2. The average composition of the PVT samples for reservoir R1 and R2 is given in [Table 2](#). These early time samples have good lateral and vertical coverage.

Table 2—Average composition of hydrocarbons for the two reservoirs

Component	Component Name	Reservoir R1	Reservoir R2
Nitrogen	N2	0.47	0.23
Carbon Dioxide	CO2	4.14	3.98
Hydrogen Sulphide	H2S	2.34	2.31
Methane	C1	71.04	70.22
Ethane	C2	6.45	6.57
Propane	C3	3.52	3.66
i-Butane	iC4	0.86	0.94
n-Butane	nC4	1.73	2.00
i-Pentane	iC5	0.82	0.97
n-Pentane	nC5	0.94	1.17
Hexane	C6	1.30	1.55
Heptane	C7	1.32	1.53
Octanes	C8	1.28	1.43
Nonanes	C9	1.01	1.16
Decanes	C10	0.82	0.80
Undecanes	C11	0.52	0.46
Dodecane Plus	C12+	1.45	1.08

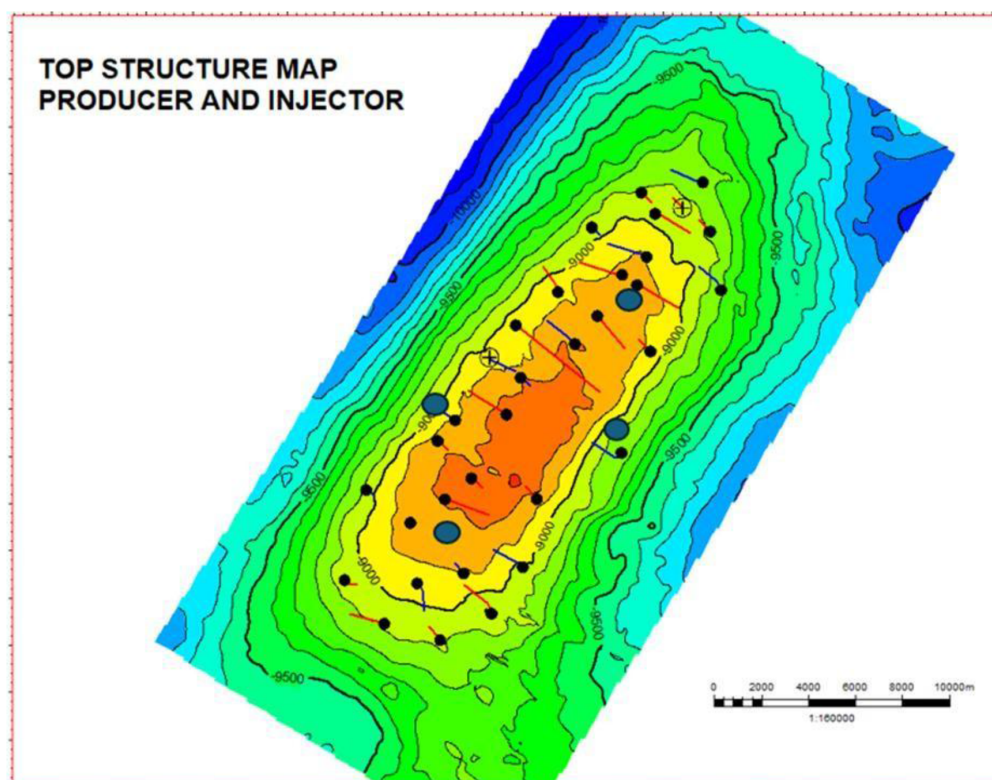


Figure 1—The top structure map of reservoir R1 with the the wells and the location of initial reservoir samples overlaid

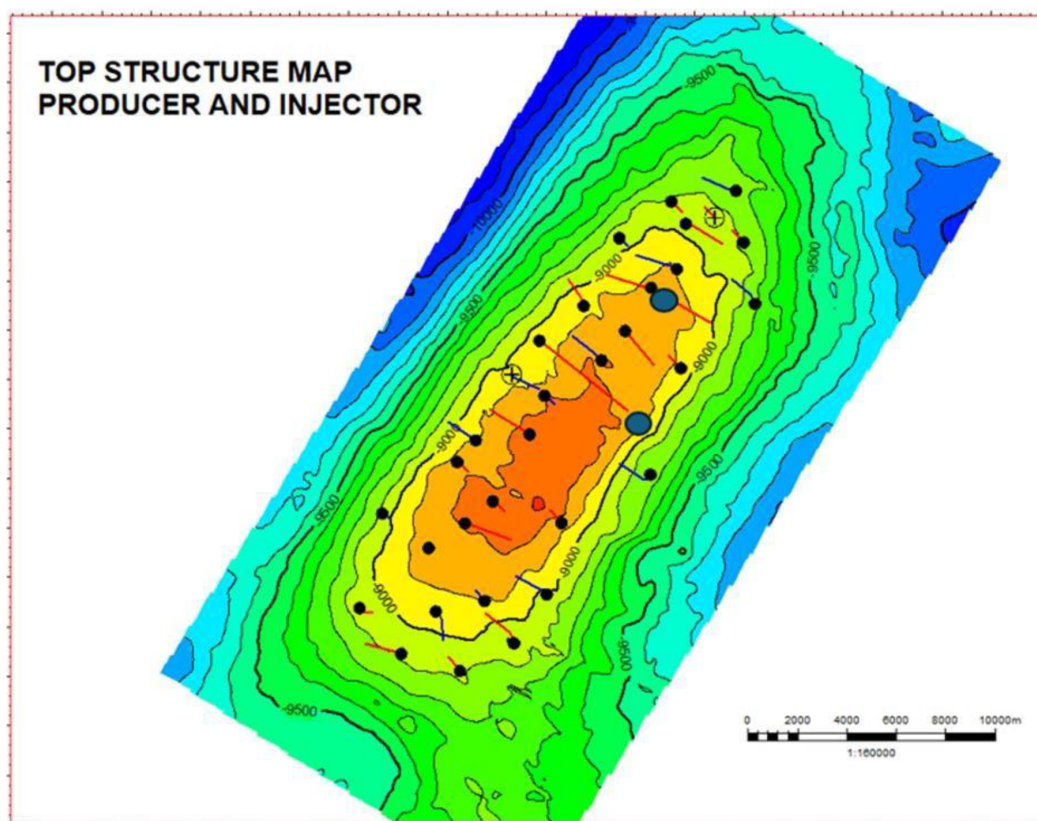


Figure 2—The top structure map of reservoir R2 with the the wells and the location of initial reservoir samples overlaid

Both reservoirs are gas bearing with high condensate to gas ratio. The initial condensate gas ratio ranges from 120 to 140 STB/MMSCF for both reservoirs. The condensate gravity is in the range of 53 to 55 °API. For the acid gas concentration, the CO₂ concentration is relatively uniform but H₂S has a tendency of increasing concentration toward the south. The high amount of H₂S and CO₂ make conventional processing of the produced gas challenging.

Field Development Plan

The developed gas reservoirs, R1 and R2 have been on production since year 2000 under gas recycling mode to maximize condensate recovery. Gas recycling with a line drive injector-producer pattern perpendicular to the field's long-axis was implemented as part of the field development plan. Partial Injection will be maintained with 75% recycle ratio until the year 2041. The current gas recycling operation will be followed by gas blowdown starting in the year 2042.

There are currently 39 gas producers and 33 gas injectors in the field. All producers are single strings, 22 of which are in R1 reservoir while the remaining 17 are in R2 reservoir. In the injectors case, there are 5 dual string wells in the field injecting in both R1 and R2. 14 single string wells in R1 reservoir and 9 single strings wells in R2 reservoir.

There are 4 gas gathering stations known as Gas Remote Stations (GRS 1 to 4) where natural gas from each production well is transferred to respective GRS's through a dedicated production flow line. From the GRSs, the condensate rich natural gas is transported through a dedicated production trunkline to the Receiver Facility (RF) and passes it to the gas processing facility. At the gas processing facility, the condensate is stabilized and separated from the production stream. The remaining sour gas is sent to the acid gas removal unit (AGRU). In this unit the acid gas H₂S and Carbon dioxide (CO₂) are removed from the feed gas by scrubbing with a solvent such as glycol. The sweet gas after the separation of the sour gas components is sent to a dehydration unit, chilling and then final to a natural gas liquids (NGL) separations unit. The NGL unit is designed to remove all molecules heavier than C₁. The lean gas output from the NGL which consists mostly of CH₄ is combined with the sour gas components (H₂S and CO₂) separated from the AGRU for re-injection back into the reservoir. This combined gas is first dehydrated, re-compressed and then transferred via 3 trunk lines for injection as lean gas back into the formations. [Figure 3](#) shows the process flow diagram for the existing facilities unit showing all the units described above.

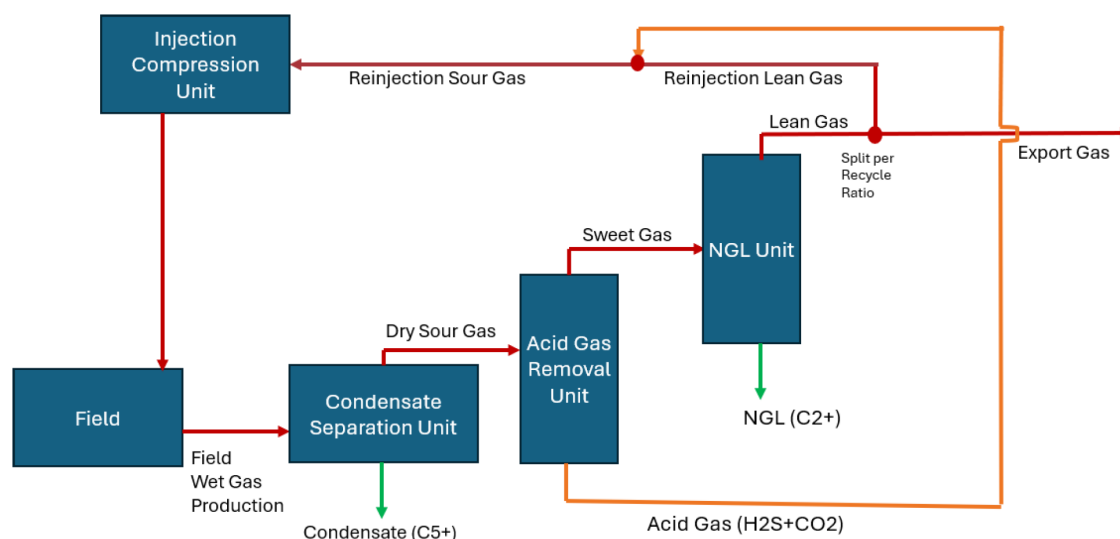


Figure 3—Simplified Facilities Flow Diagram for the subject field

Dynamic Model and History Match

The current average reservoir pressure in R1 is 4035 psia while the reservoir pressure in R2 reservoir which is tighter and more heterogeneous of the two reservoirs is 4,400 psia respectively.

A dynamic compositional model was used to generate the production and injection profiles. The model was history- matched from May 2000 to December 2024. The production and injection strings were run under gas rate control and were calibrated with allocated gas, condensate, bottom hole shut in pressure and flowing tubing head pressure. The overall gas, condensate, CGR, gas injection rate and the pressure matches for the field are shown in Figures 4 to 7.

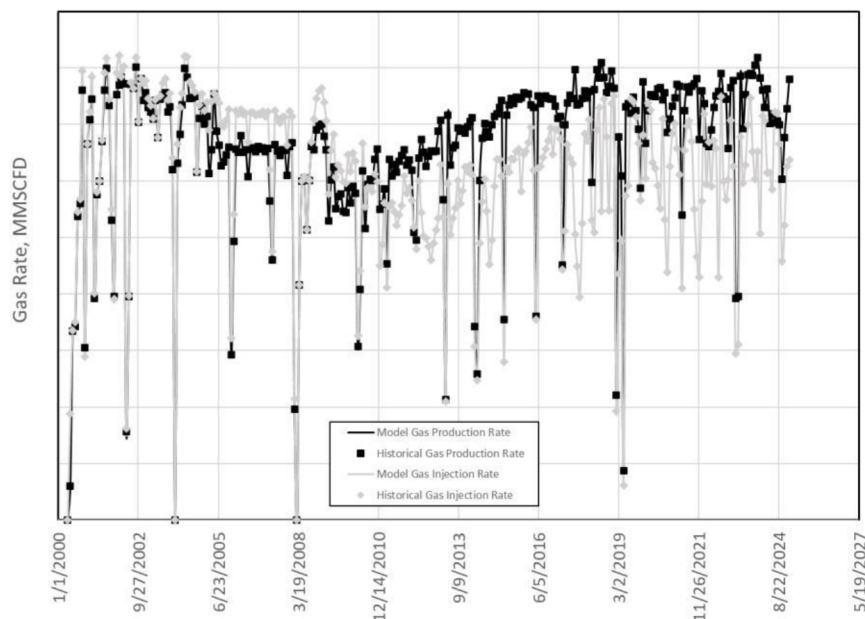


Figure 4—Gas rate match of field historical production and injection to the calibrated dynamic simulation model

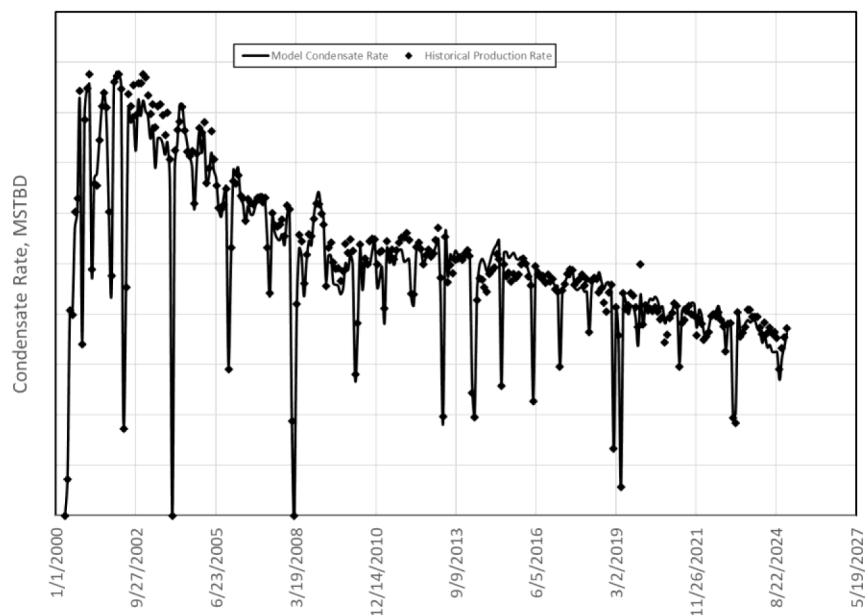


Figure 5—Condensate rate match of field historical production to the calibrated dynamic simulation model

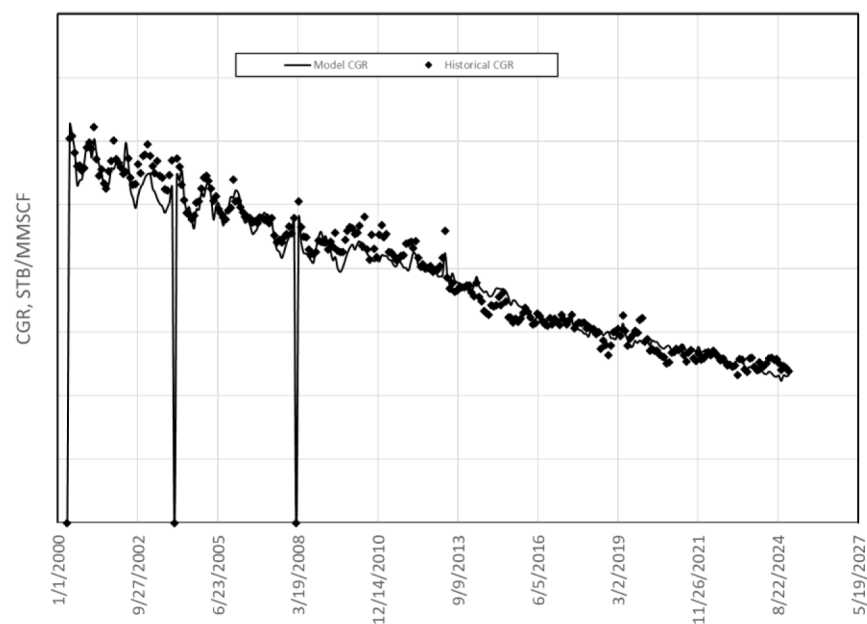


Figure 6—CGR match of field historical production to the calibrated dynamic simulation model

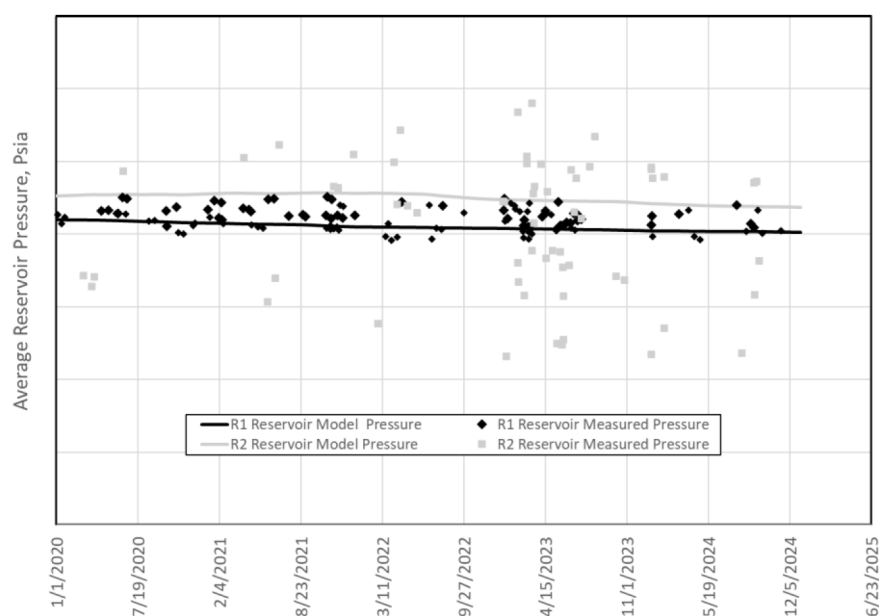


Figure 7—Reservoir pressure match of field measured data to the calibrated dynamic simulation model

The history-match to field data was achieved mainly by modifying the permeability to certain level and proper quality control has been done such as comparing kh well in the model vs kh from the well testing, well production and well pressure history. In general, good to excellent match was achieved in both the reservoirs.

Installation of SRU/CCU and Field Redevelopment Plan

The current field redevelopment plan involves installation of a Sulphur Recovery Unit (SRU) and Carbon Capture Unit (CCU) as a way to monetize the sulphur and CO₂ production from the field. The SRU/CCU is expected to be installed and commissioned at the processing plant by end of year 2028 (Q4-2028). This would unlock anticipated production ramp-up of approximately 12 % from the current production levels. It is expected that the production ramp-up will enhance injection by a corresponding amount as the field is

operated at a recycle-ratio of 75 % as mentioned earlier. However, the installation of the SRU would limit the H₂S and CO₂ to 2 mol% and 3-4mol% respectively in the injection stream and gradually drop lower as will be shown in the simulation model forecasts. The production stream H₂S and CO₂ mole fractions are also expected to decrease with the installation of the SRU/CCU. A 2nd SRU installation is to follow in 2042, aligning with the blow-down year. Figure 8 shows the updated process flow diagram with the installation of the SRU and CCU in the existing facilities. The sulphur recovery and carbon capture utilization strategy are given in detail in the following sections.

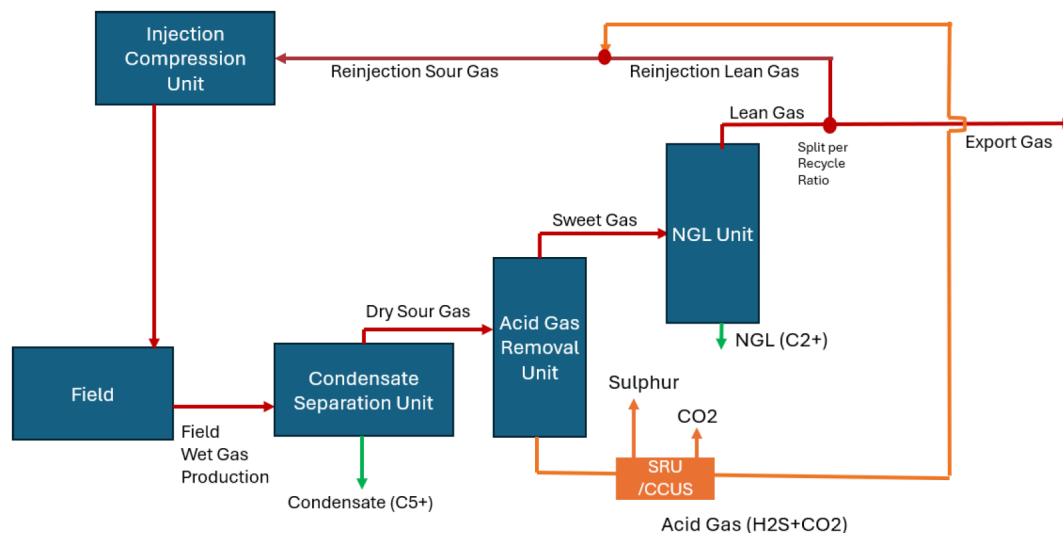


Figure 8—Simplified Facilities Flow Diagram for the subject field with the installation of the SRU/CCUS system

The forecast prediction model was set up at the field level targeting the mandated field gas production rate of wet gas and ramp up by 12 % in Jan 2029 as per the field redevelopment plan discussed in the previous section. Figure 9 and Figure 10 show the forecast of the gas production, injection and the condensate production streams with and without the installation of the SRU/CCUS. As can be seen, the installation of SRU/CCUS unlocks the potential to increase the production by 12% C associated increase in injection streams at a field level. Moreover, as can be seen in Figures 11 and 12, the installation of the SRU and CCUS will result in the concentrations of the H₂S and CO₂ to decrease both in the injection and the production streams. Figure 11 shows the concentration (mole fraction) of H₂S in the injection and the production streams for a scenario with SRU installation and another scenario without the SRU installation. As can be seen in this figure, in the scenario without the installation of the SRU, the H₂S mole fraction in the injection stream as well as in the production stream starts to rise in 2029 when the production is ramped up by 12 % as was discussed earlier. The H₂S mole fractions can reach up to 5.7 % in the injection stream and 4.12 % in the production stream by the end of the recycling injection period and these high H₂S concentrations can be detrimental to the downstream plants and facilities. However, in the scenario with the installation of the SRU, the H₂S mole fraction drops drastically post 2029 with the commissioning of the SRU and stays on a decreasing trend until it becomes zero in year 2040. Accordingly, the H₂S mole fraction in the production stream also decreases till the blowdown year of 2042. There is a slight increase in H₂S mole fraction in the production stream after 2042 as the reservoir is put on blowdown and any H₂S remaining in the reservoir is now produced back up. However, the H₂S mole fractions at these late times in the SRU installation scenario are very low and do not impact any of the surface pipelines or downstream processing facilities. Similarly, Figure 12 shows the concentration of the CO₂ in the injection and the production streams for a scenario with CCU installation and another scenario without the

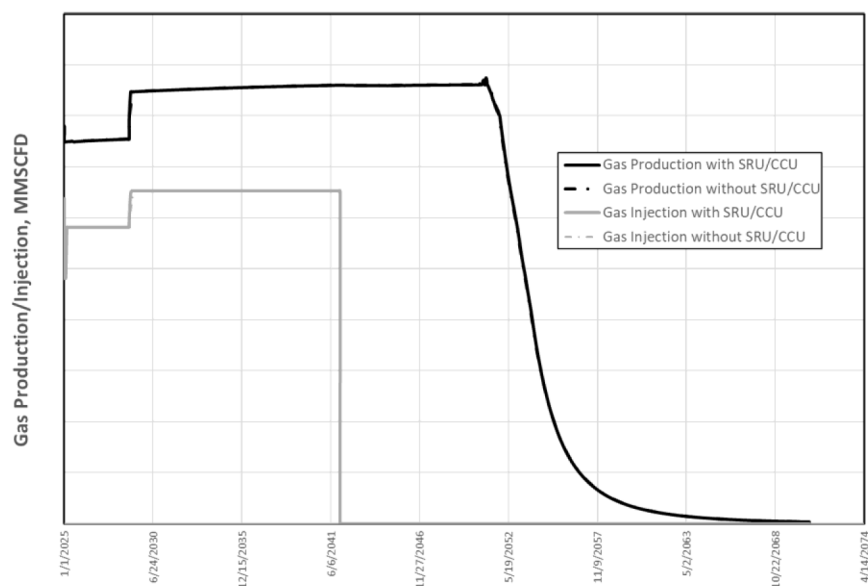


Figure 9—Gas rate forecast for production and injection from the calibrated dynamic simulation model with and without the SRU/CCU

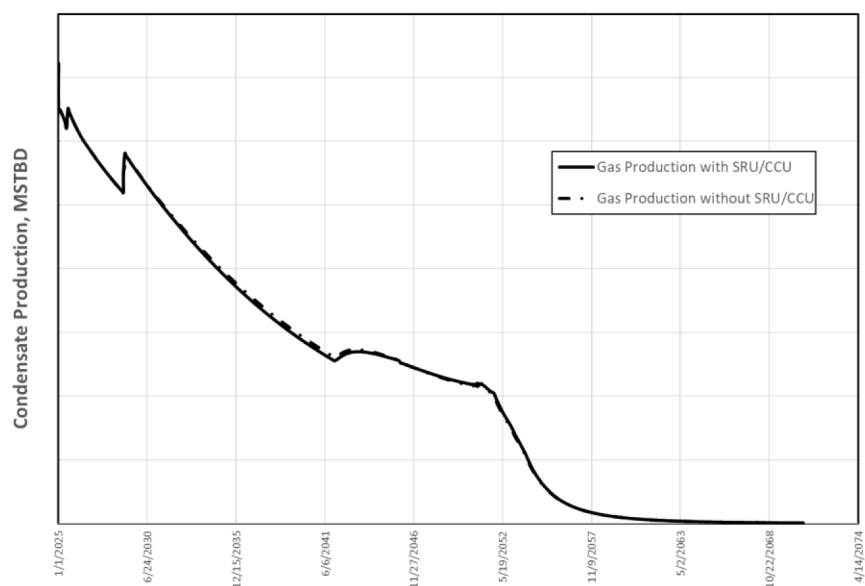


Figure 10—Condensate production rate forecast from the calibrated dynamic simulation model with and without the SRU/CCU

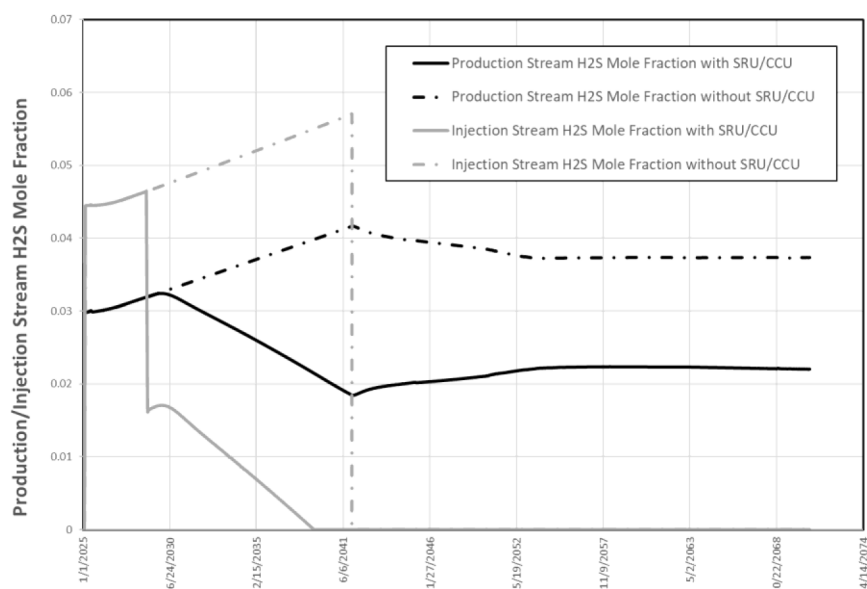


Figure 11—H₂S mole fraction from the calibrated dynamic simulation model with and without the SRU/CCUS

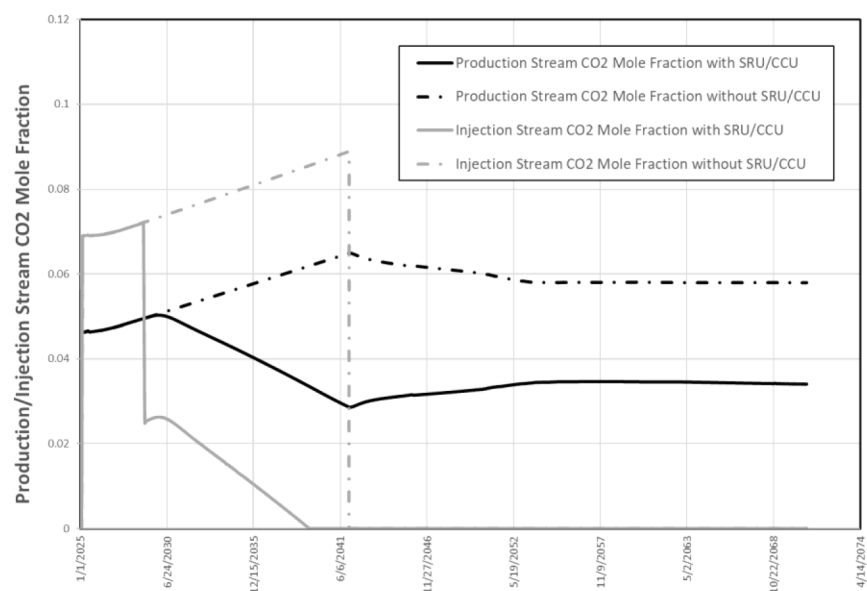


Figure 12—CO₂ mole fraction from the calibrated dynamic simulation model with and without the SRU/CCUS

CCU installation. The trends of the CO₂ mole fraction in the two scenarios in the two streams (production and injection) are very similar to the trends for the H₂S in the two scenarios. The CO₂ mole fractions can reach up to 8.8 % in the injection stream and 6.6 % in the production stream by the end of the recycling injection period and these high CO₂ concentrations can be detrimental to the downstream plants and facilities. However, in the scenario with the installation of the CCU, the CO₂ mole fraction drops drastically post 2029 with the commissioning of the CCU and stays on a decreasing trend until it becomes zero in year 2040. Accordingly, the CO₂ mole fraction in the production stream also decreases till the blowdown year of 2042. There is a slight increase in CO₂ mole fraction the production stream after 2042 as the reservoir is put on blowdown and any CO₂ remaining in the reservoir is now produced back up. However, the CO₂ mole fractions at these late times in the CCU installation scenario are very low and do not impact any of the surface pipelines or downstream processing facilities.

With the current configuration of the SRU and CCU the H₂S and the CO₂ in the injection stream are estimated to drop to zero by year 2040 with corresponding drop in the production stream as well, as per the current simulation forecast model. This enables enhanced production from the field while operating the facilities safely with regards to the acid gas components.

Sulphur Recovery and Carbon Capture Utilization Strategy

The integration of a Sulphur Recovery Unit (SRU) and a Carbon Capture, Utilization, and Storage (CCUS) system within the field's processing architecture is a pivotal enhancement aligned with ADNOC Gas's broader decarbonization and monetization strategy. These facilities are designed not only to mitigate environmental and operational risks associated with increasing acid gas reinjection but also to unlock value through additional revenue streams and increased oil recovery from nearby fields.

Sulphur Recovery Unit (SRU)

The SRU is designed to process H₂S-rich streams extracted from the Acid Gas Removal Unit (AGRU), converting hydrogen sulphide into elemental sulphur via a modified Claus process (Ref.: 1 C 2). This process involves thermal and catalytic conversion steps that allow recovery of up to 98% of sulphur content in the feed gas.

- **Thermal Stage:** Acid gas is combusted in a thermal reactor at 1300–1400°C, partially oxidizing H₂S to SO₂. This reaction is:
$$2\text{H}_2\text{S} + 3\text{O}_2 \rightarrow 2\text{SO}_2 + 2\text{H}_2\text{O}$$
- **Catalytic Stages:** The remaining H₂S and SO₂ react over alumina/titania-based catalysts at 250–350°C to form elemental sulphur:
$$2\text{H}_2\text{S} + \text{SO}_2 \rightarrow 3\text{S} + 2\text{H}_2\text{O}$$

To enhance Sulphur recovery efficiency (>98%), a **Tail Gas Treatment Unit (TGTU)** using amine-based solvent is incorporated, allowing for near-complete H₂S removal from vent streams and reducing SO_x emissions below regulatory limits.

Design Features

The design capacity of the unit is 15 tons/day of elemental sulphur, with provisions for expansion in future phases. This recovered sulphur is planned to be granulated and marketed, contributing to ADNOC Gas's non-hydrocarbon revenue generation portfolio.

Operational Impact

From an operational standpoint, implementation of the SRU mitigates the corrosive and toxic impact of H₂S on injection and surface facilities, ensuring compliance with integrity limits of pipelines and downhole tubulars. Moreover, reducing the H₂S content in the injection stream down to 2 mol% (and ultimately 0 mol% by 2041) supports safe operations throughout the life of the field and enables the processing plant to remain within its environmental compliance thresholds.

Carbon Capture, Utilization, and Storage (CCUS)

Parallel to sulphur recovery, a dedicated CO₂ capture and injection system is being developed. CO₂ stripped in the AGRU is routed to a post-treatment section using DMEA(Dimethylethanolamine)-based solvent technology, which offers high selectivity and thermal stability for CO₂ capture. The dehydrated CO₂ stream is then compressed and transported via pipeline to the Bab Far North platform, where it is injected into depleted oil reservoirs.

EOR Mechanism

This targeted CO₂ injection serves as an enhanced oil recovery (EOR) technique, enabling pressure maintenance and improved sweep efficiency. From an economic standpoint, diverting CO₂ for EOR instead of full reinjection significantly improves the project's value proposition. The combined benefit of CO₂ monetization through incremental oil recovery and elemental sulphur sales offers a dual revenue stream, while also enabling ADNOC Gas to meet its GHG emission intensity reduction targets.

Environmental Benefit

- Avoids reinjection of high-CO₂ acid gas, reducing wellhead souring risks.
- Sequesters up to **~1.2 million tons/year of CO₂**, directly supporting ADNOC's GHG emission intensity reduction target of 25% by 2030.

Strategic Integration and phased Execution

The SRU/CCUS system is planned to be phased into two key stages:

Phase 1 (2028–2041): Handles full acid gas load from current production. Targets reduction of H₂S and CO₂ in the injection stream from baseline values (~4–5 mol%) to near-zero.

Phase 2 (2042 onward): Supports the gas blowdown phase by removing residual acid gas and maintaining export gas specifications as field transitions out of recycle mode.

Simulation forecasts predict H₂S in the injection stream will decline to <0.1 mol% by 2041, with CO₂ approaching zero by 2042, ensuring compliance with long-term field development and injection system integrity limits.

Summary

In summary, the planned installation of the SRU and the CCU will gradually lower the concentrations of the acid gas components in the injection and production streams which will help in safe operation of the facilities. This implementation also helps in unlocking the production potential of the reservoir by enabling the ramp-up of the production from the field by approximately 10- 12 %. The elemental sulphur produced would generate an additional revenue stream. The captured CO₂ will be sent to a nearby oil field for enhanced oil recovery, adding to the existing revenue streams.

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